Examples

1 Distance between two slits is 0.1 mm and the width of the fringes formed on the screen is 5 mm. If the distance between screen and slit is one meter calculate the wavelength of the light used. **[NLJIET]**

1) $d = 0.1\,mm$

$\beta = 5\,mm$

$D = 1\,m$

$\lambda = ?$

$$\beta = \frac{\lambda D}{d}$$

$$\therefore \lambda = \frac{\beta d}{D}$$

$$= \frac{5 \times 10^{-3} \times 0.1 \times 10^{-3}}{1}$$

$$= 5 \times 10^{-7}\,m$$

$$\boxed{\lambda = 500\,nm}$$

2 In Newton's ring experiment, the diameter of a certain bright ring is o.65 cm, and that of tenth ring beyond is 0.95 cm. If λ= 6000 A$^{\circ}$, calculate radius of curvature of a convex lens surface in contact with the glass plate. **[NLJIET]**

2. $D_n = 0.65\,cm$

$D_{n+10} = 0.95\,cm$

$\lambda = 6000\,A^{\circ} = 6000 \times 10^{-10}\,m$

$R = ?$

$$R = \frac{D_{n+p}^2 - D_n^2}{4 P \lambda}$$

$P = 10$

$$R = \frac{(0.95 \times 10^{-2})^2 - (0.65 \times 10^{-2})^2}{4 \times 10 \times 6000 \times 10^{-10}}$$

$$= \frac{0.9025 \times 10^{-4} - 0.4225 \times 10^{-4}}{2.4 \times 10^{5} \times 10^{-10}}$$

$$= \frac{0.48 \times 10^{-4}}{2.4 \times 10^{-5}}$$

$$= 0.2 \times 10$$

$$\boxed{R = 2\,m}$$

3	Newton's rings are observed in reflected light of λ = 5900 A°. The diameter of the 5th dark ring is 0.4 cm. Find the radius of curvature of the lens and the thickness of the air film. **[NLJIET]**

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3. $\lambda = 5900\ A^\circ$

$D_5 = 0.4\ cm$

$R = ?$

$t = ?$

$$D_n^2 = 4n\lambda R$$

$$D_5^2 = 4 \times 5 \times 5900 \times 10^{-10} \times R$$

$$\rightarrow \quad R = \frac{(0.4 \times 10^{-2})^2}{1.18 \times 10^{5} \times 10^{-10}}$$

$$= \frac{0.16 \times 10^{-4}}{1.18 \times 10^{-5}}$$

$$= 0.1355 \times 10$$

$$\boxed{R = 1.355 m}$$

$$\rightarrow \quad 2t = n\lambda$$

$$t = \frac{5 \times 5900 \times 10^{-10}}{2}$$

$$t = 1.4750 \times 10^{-6}$$

$$\boxed{t = 1.475\ \mu m}$$

4	If the diameter of 5th and 14th dark rings are 0.525 cm and 0.900 cm respectively in a Newton's rings experiment, find the diameter of 23rd ring. **[NLJIET]**

4. $D_5 = 0.525$ cm

$D_{14} = 0.900$ cm

$D_{23} = ?$

$$\rightarrow \quad R = \frac{D^2_{n+p} - D^2_n}{4P\lambda}$$

$$R = \frac{D^2_{14} - D^2_5}{4 \times 9 \times \lambda}$$

$$R\lambda = \frac{(0.900 \times 10^{-2})^2 - (0.525 \times 10^{-2})^2}{36}$$

$$= \frac{0.81 \times 10^{-4} - 0.2756 \times 10^{-4}}{36}$$

$$= \frac{0.5344 \times 10^{-4}}{36}$$

$$R\lambda = 0.0148 \times 10^{-4}$$

$$\rightarrow \quad D^2_n = 4n\lambda R$$

$$D^2_{23} = 4 \times 23 \times 0.0148 \times 10^{-4}$$

$$D^2_{23} = 1.3616 \times 10^{-4}$$

$$\boxed{D_{23} = 1.1668 \times 10^{-2}} \text{ m.}$$

5	In Na yellow doublet has wavelengths 5890 A˚ and 5896 A˚. What should be the R.P. of a grating to resolve these lines? **[NLJIET]**

5. $\lambda_1 = 5890$ A°

$\lambda_2 = 5896$ A°

R.P. = ?

$$R.P = \frac{\lambda}{d\lambda}$$

$$= \frac{5893}{6}$$

$$\boxed{R.P = 982}$$

$$\lambda = \frac{\lambda_1 + \lambda_2}{2}$$

$$= \frac{5896 + 5890}{2}$$

$$= 5893 \text{ A}^\circ$$

$$d\lambda = \lambda_2 - \lambda_1$$

$$= 6 \text{ A}^\circ$$

6	Calculate the minimum number of lines in a grating which will just resolve the lines of wavelength 5890 Å and 5896 Å in the second order. **(Jan-2025-New) [NLJIET]**

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$\lambda_1 = 5890 A^\circ$

$\lambda_2 = 5896 A^\circ$

$m = 2$

$N = ?$

$\lambda = \dfrac{\lambda_1 + \lambda_2}{2}$

$\lambda = 5893 A^\circ$

$d\lambda = 6 A^\circ$

$$R.P = \frac{\lambda}{d\lambda} = mN$$

$$\therefore N = \frac{\lambda}{m\, d\lambda}$$

$$= \frac{5893}{2 \times 6}$$

$$= 491.08 \approx 491$$